



Course : R-VIJETA (R-JP)

PHYSICS DPP

DAILY PRACTICE PROBLEM

REVISION DPP NO. # 04

Target : JEE (Advanced) 2023

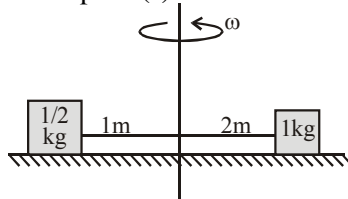
DPP Syllabus : CIRCULAR MOTION, CENTRE OF MASS AND RIGID BODY DYNAMICS DPP NO. 04

Maximum Marks : 170

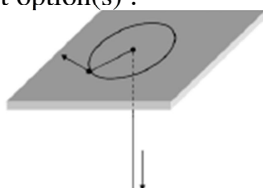
Total Time : 160 Min.

One or more than one may be correct option ('-1' Negative marking) Q.1 to 18; (3 Marks; 2 Min.) [54, 36]
 Comprehension Choice ('-1' Negative marking) Q.19 to Q.20 (3 Marks; 2 Min.) [06, 04]
 Subjective questions ('-1' Negative marking) Q.21 to Q.40; (4 Marks; 5 Min.) [80, 100]
 Single Choice Objective ('-1' Negative marking) Q.41 to Q.44 (3 Marks; 2 Min.) [12, 08]
 Comprehension Choice ('-1' Negative marking) Q.45 to Q.50 (3 Marks; 2 Min.) [18, 12]

1. Two small blocks tied with a massless string of length 3 m are placed on a rotating table as shown. The axis of rotation is 1 m from $\frac{1}{2}$ kg mass and 2 m from 1 kg mass. The angular speed $\omega = 1$ rad/s and the blocks do not slip. Surface below 1 kg block is smooth and that below $\frac{1}{2}$ kg block is rough (friction coefficient $\mu = 0.6$). Choose the correct option(s).



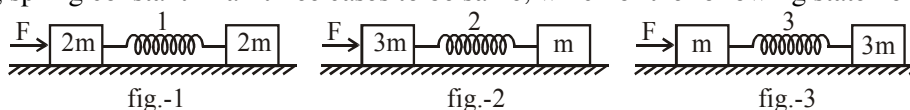
- (A) Tension in the string is 2 N (B) Tension in the string is 1 N
 (C) Friction force on $\frac{1}{2}$ kg block is $\frac{3}{2}$ N (D) Resultant force on the system is zero.
2. A mass is attached to the end of a string. The mass moves on a frictionless table, the string passes through a hole in the table, under which some one is pulling on the string to make it taut at all times. Initially, the mass moves in a circle of radius R with kinetic energy E_0 . The string is slowly pulled, until the radius of the circle is halved. Choose the correct option(s) :-



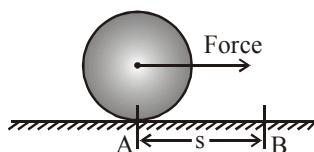
- (A) Initially the tension in string is $\frac{2E_0}{R}$.
 (B) Initially the tension in string is $\frac{L^2}{mR^3}$, where L is angular momentum of the particle about center.
 (C) Work done to reduce the radius half is $\frac{3L^2}{2mR^2}$, where L is angular momentum about center.
 (D) Work done to reduce the radius half is $3E_0$.

3. In a one-dimensional inelastic collision between two identical particles A and B of equal masses, initially B is stationary and A has momentum P before impact. During impact, B gives impulse of magnitude J to A, then Choose the correct option(s) :
- (A) The total momentum of the "A plus B" system is P before and after the impact and $(P-J)$ during the impact.
- (B) During the impact, A gives impulse of magnitude J to B.
- (C) The coefficient of restitution is $\frac{2J}{P} - 1$.
- (D) The coefficient of restitution is $\frac{J}{P} + 1$.

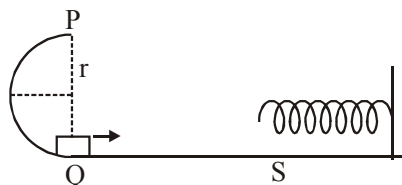
4. In each of three figures shown, two blocks are connected by a light spring and the system is placed on smooth horizontal surface. A constant horizontal force of magnitude F is applied to left block as shown. Assuming spring constant in all three cases to be same, which of the following statements is/are true?



- (A) maximum compression in spring 1 is greater than that in spring 2.
- (B) maximum compression in spring 3 is greater than that in spring 1.
- (C) maximum compression in spring 3 is greater than that in spring 2.
- (D) maximum compression in all springs is equal.
5. A disc of circumference 's' is at rest at point A on a horizontal surface when a constant horizontal force begins to act on its centre. Between A and B there is sufficient friction to prevent slipping, and the surface is smooth to the right of B. $AB = s$. The disc moves from A to B in time T . Then to the right of B :

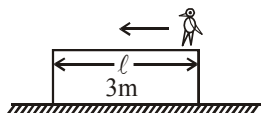


- (A) the angular acceleration of the disc will disappear, linear acceleration will remain unchanged.
- (B) linear acceleration of the disc will increase.
- (C) the disc will make one rotation in time $T/2$.
- (D) the disc will cover a distance greater than 's' in further time T .
6. The circular vertical section of the fixed track shown is smooth with radius $r = 0.9$ cm and the horizontal straight section is rough with $\mu = 0.1$. A block of mass 1 kg is placed at point 'Q' and given a horizontal velocity of m/s towards the spring. Distance $QS = 40$ cm and maximum compression in the spring is 10 cm during the motion ($g = 10 \text{ m/s}^2$) :

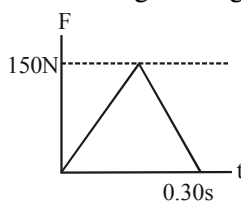


- (A) The force constant of the spring is 200 N/m.
- (B) The velocity with which block returns to point 'Q' is 1 m/s.
- (C) At point P its velocity will be 0.8 m/s.
- (D) At point P, the normal reaction on the block is less than 55 N.

7. Two particles move on a circular path (one just inside and the other just outside) with angular velocities ω and 5ω starting from the same point. Then :
- (A) they cross each other at regular intervals of time $\frac{2\pi}{4\omega}$ when their angular velocities are oppositely directed.
- (B) they cross each other at points on the path subtending an angle of 60° at the centre if their angular velocities are oppositely directed.
- (C) they cross at intervals of time $\frac{\pi}{3\omega}$ if their angular velocities are oppositely directed.
- (D) they cross each other at points on the path subtending 90° at the centre if their angular velocities are in the same sense.
8. A penguin of mass m stands at the right edge of a sled of mass $3m$ and length ℓ . The sled lies on frictionless ice. The penguin starts moving towards left, reaches the left end and jumps with a velocity u at an angle θ relative to ground. (Neglect the height of the sled) :

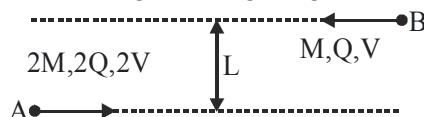


- (A) Till the penguin reaches the left end, the sled is displaced by $\frac{\ell}{4}$.
- (B) Till the penguin reaches the left end, the sled is displaced by $\frac{\ell}{3}$.
- (C) After jumping, it will fall on the ground at a distance $\frac{4}{3} \frac{u^2 \sin 2\theta}{g}$ from the left end of the sled.
- (D) After jumping, it will fall on the ground at a distance $\frac{3}{4} \frac{u^2 \sin 2\theta}{g}$ from the left end of the sled.
9. Two bodies A and B of masses 5.00 kg and 10.0 kg respectively are moving in opposite directions with velocities 4.00 m/s and 0.50 m/s respectively make head-on collision in free space. The force of their mutual interaction during collision varies according to the given graph.

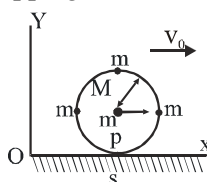


- (A) Period of deformation is 0.20 s.
- (B) Coefficient of restitution is 0.50.
- (C) Body A will move with velocity 0.50 m/s in the original direction.
- (D) Body B will move with velocity 1.75 m/s in the original direction.

10. Two particles A and B approach each other from very large distance as shown in the figure. Their masses, charges and speeds are mentioned in the figure. (Neglect gravitational force) :

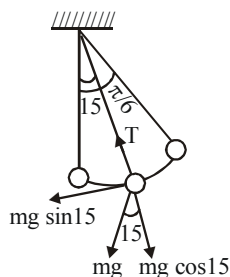


- (A) If the least separation between the particles is b , the speed of M in center of mass frame when particles are closest, is $2LV/b$.
- (B) If the least separation between the particles is b , the speed of M in center of mass frame when particles are closest, is LV/b .
- (C) If the least separation between the particles is b , the speed of $2M$ in center of mass frame when particles are closest, is $2LV/b$.
- (D) If the least separation between the particles is b , the speed of $2M$ in center of mass frame when particles are closest, is LV/b .
11. A uniform circular disc of mass M and radius R has four particles (each of mass m) rigidly attached to it. The disc rolls on the surface S without slipping. At the instant shown, the centre of the disc has velocity v_0 :

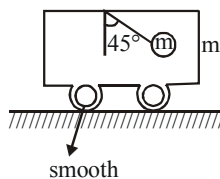


- (A) The kinetic energy of the disc-particle assembly is $\left(\frac{9m}{2} + \frac{3M}{4}\right)v_0^2$.
- (B) The velocity of the centre of mass of the assembly is $\left(\frac{M+5m}{M+4m}\right)v_0\hat{i}$.
- (C) The angular momentum of the assembly about the point of contact P is $\frac{3}{2}(M+6m)Rv_0(-\hat{k})$.
- (D) The maximum speed of a point on the disc with respect to S is $2v_0$.
12. A thin rod of mass m and length ℓ is free to rotate on a smooth horizontal plane about its one fixed end. When it is at rest, it receives a horizontal impulse J at its other end, at angle of 37° with the length. Immediately after impact :
- (A) Angular momentum of the rod is $0.6J\ell$.
- (B) Angular velocity of the rod is $\frac{1.8J}{m\ell}$.
- (C) Kinetic energy of the rod is $\frac{0.54J^2}{m}$.
- (D) Linear velocity of the centre of mass of the rod is $\frac{9}{10}\frac{J}{m}$.

13. A simple pendulum of length 2 m with a bob of mass M oscillates with an angular amplitude of $\frac{\pi}{6}$ radians, then (use $\sqrt{g} = \pi$) :

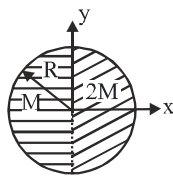


- (A) tension in the string is $mg \cos 15^\circ$ at angular displacement of 15° .
 (B) rate of change of speed at angular displacement of 15° is $g \sin 15^\circ$.
 (C) tension in the string is greater than $mg \cos 15^\circ$ at angular displacement of 15° .
 (D) frequency of oscillation is 0.5 sec^{-1} .
14. A heavy particle is tied to the end A of a string of length 1.6 m. Its other end O is fixed. It revolves as a conical pendulum with the string making 60° with the vertical. Then ($g = 9.8 \text{ m/s}^2$) :
- (A) its period of revolution is $\frac{4\pi}{7} \text{ sec}$.
 (B) the tension in the string is doubled the weight of the particle.
 (C) the velocity of the particle = $2.8\sqrt{3} \text{ m/s}$.
 (D) the centripetal acceleration of the particle is $9.8\sqrt{3} \text{ m/s}^2$.
15. A cart of mass m is placed on a smooth horizontal surface. A pendulum of mass m is released from rest as shown. Then :



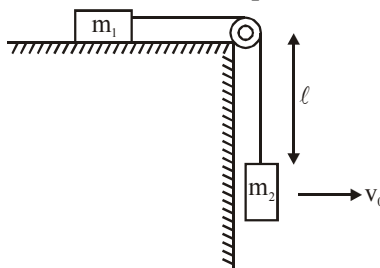
- (A) velocity of the cart just after release is $V \rightarrow 0$.
 (B) acceleration of the cart just after release is $g/3$.
 (C) velocity of the pendulum relative to the cart just after release is $V_{\text{rel}} \rightarrow 0$.
 (D) acceleration of pendulum relative to the cart just after release is $g/3$.

16. A disk of mass '3 M' and radius 'R' is made by two semicircular disks of mass 'M' and '2 M' (as shown in figure) if I_x , I_y and I_z are moment of inertia about x,y and z axis respectively, then :

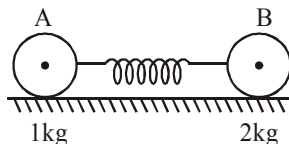


- (A) $I_x = \frac{3MR^2}{4}$ (B) $I_x = \frac{3}{2}MR^2$ (C) $I_z = \frac{3}{2}MR^2$ (D) $I_z = 3MR^2$

17. In the arrangement shown in the figure, initially the block of mass m_1 is in limiting equilibrium. The coefficient of friction between the block of mass m_1 and floor is μ . Now the particle of mass m_2 is given a horizontal velocity v_0 . The initial distance of the block of mass m_2 from the pulley is ℓ . The masses of pulley and threads are negligible. Choose the correct option(s) :



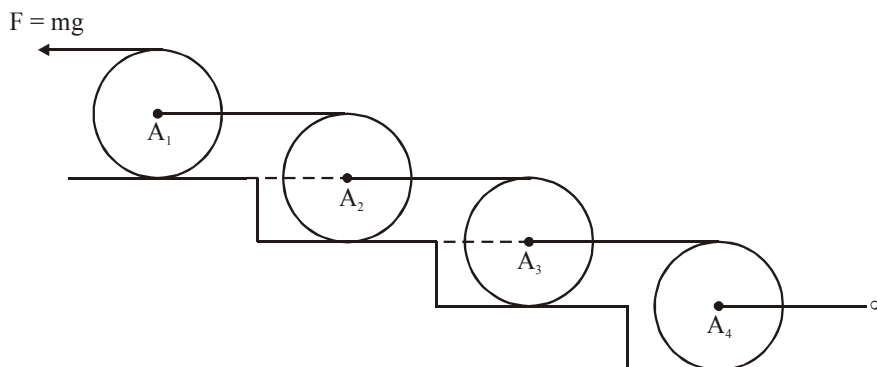
- (A) Acceleration of m_1 just after the velocity given is $\left(\frac{\mu}{1+\mu}\right)\frac{v_0^2}{\ell}$.
 (B) Acceleration of m_1 just after the velocity given is $\left(\frac{\mu}{1+\mu}\right)\frac{v_0^2}{2\ell}$.
 (C) Radius of curvature of path of m_2 just after the velocity given is $\ell(1+\mu)$.
 (D) Radius of curvature of path of m_2 just after the velocity given is $2\ell(1+\mu)$.
18. Two uniform disc A and B having masses 1 kg and 2 kg and having equal radius are connected (at centres) by a massless spring of spring constant $k = 1000 \text{ N/m}$ and kept on sufficiently rough horizontal surface. Initially system is released from rest and spring is compressed by 10 cm. Then :



- (A) friction force on A and B are always equal and opposite.
 (B) friction force on A and B depends on masses.
 (C) speed of disc A is $\sqrt{\frac{40}{9}}$ m/s, when spring is in its natural length.
 (D) frictional force on disc A is zero, when spring is in its natural length.

Paragraph for Question Nos. 19 to 20

Consider the given arrangement of infinite uniform discs $A_1, A_2, A_3, \dots, A_n$, each of mass m and radius R . Each disc is kept on a stair. Height of each stair is R . Strings are wrapped around discs whose another end is attached to centre of discs as shown. All surfaces are sufficiently rough, so that there is no slipping anywhere. A tangential horizontal force, equal to mg is applied on the disc A_1 . Whole arrangement is shown in figure. Answer the following two questions.



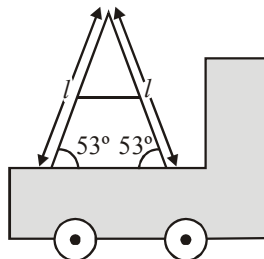
19. Choose the correct option(s) -

- (A) Acceleration of centre of mass of A_1 is g .
- (B) Acceleration of centre of mass of A_6 is $\frac{g}{32}$.
- (C) Angular acceleration of centre of mass of A_7 is $\frac{g}{64R}$.
- (D) Angular acceleration of centre of mass of A_5 is $\frac{g}{16R}$.

20. Choose the correct option(s) -

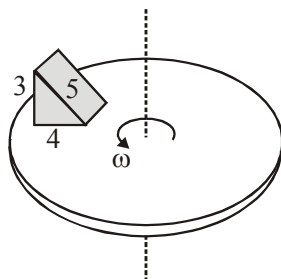
- (A) Tension in the string connecting A_1 and A_2 is $\frac{mg}{2}$.
- (B) Tension in the string connecting A_5 and A_6 is $\frac{mg}{32}$.
- (C) Frictional force between A_8 and stair is $\frac{mg}{256}$.
- (D) Frictional force between A_3 and stair is $\frac{mg}{8}$.

21. A V shaped ladder is kept on the truck as shown.

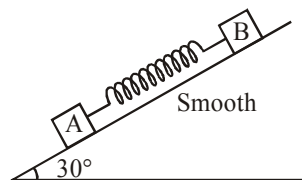


Friction between the truck and the ladder is sufficient enough to prevent any slipping. A rod is fixed between two arms (at their midpoints) of the ladder so that ladder doesn't collapse. This rod is massless and mass of the ladder is 'm', which is uniformly distributed throughout its length. Truck is initially at rest. Truck can accelerate with maximum acceleration 20 m/s^2 and retard with max retardation 10 m/s^2 . Maximum speed of the truck can be 60 m/s . This truck is to complete a journey of 1500 m with maintaining the ladder in same position with respect to the truck as it was before the truck started moving, starting from rest and ending at rest. Minimum possible time taken by the truck for this task is T_0 sec. Find T_0 . (Use $g = 10 \text{ m/s}^2$)

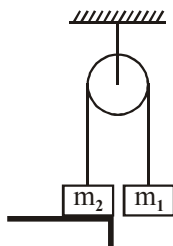
22. A 3-4-5 inclined plane is fixed to a rotating turntable. A block rests on the inclined plane and the coefficient of static friction between the inclined plane and the block is $\mu_s = \frac{1}{4}$. The block is to remain at a position 40 cm from the center of rotation of the turntable (figure). Find the minimum angular velocity ω (in rad/sec) to keep the block from sliding down the plane (toward the center) (Round off to the nearest integer).



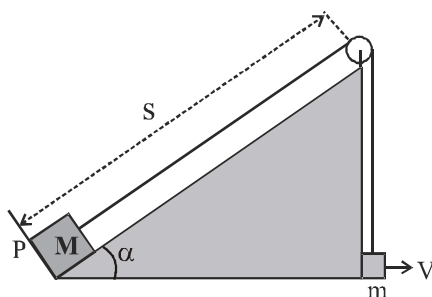
23. In given arrangement, both blocks are of same mass 2 kg . Initially spring is compressed by compression 10 cm . Spring constant is 200 N/m . Find the ratio of magnitude of the initial relative acceleration of the two blocks and acceleration of centre of mass of two blocks.



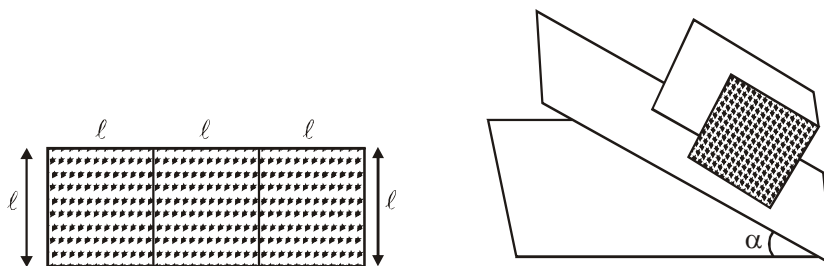
24. Two masses $m_1 = 1 \text{ kg}$ & $m_2 = 2 \text{ kg}$, are connected by a fine inelastic taut string shown. Then m_1 is raised vertically by height 'h = 20 m' and let fall. Find the the interval (in second) between m_2 leaving the table and returning to it again.



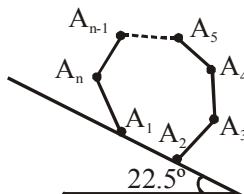
25. Consider the system as shown in figure. The body of mass $M = 3 \text{ kg}$ leans against wall P placed at the bottom of a smooth inclined plane of an angle $\alpha = 30^\circ$ and length $s = 10 \text{ m}$. This body is connected by an inelastic string passed over an ideal pulley mounted at the top of the plane to another body of mass $m = 1 \text{ kg}$ as shown. What should be horizontal velocity (in m/sec) of m so that the other the body (M) may detach from the wall.



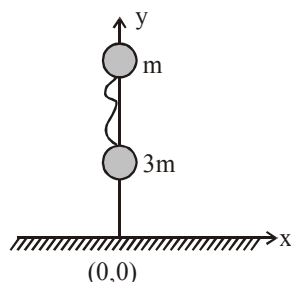
26. A cardboard strip, bent in the shape of the letter C, is put on a rough inclined plane, as shown in the figure. If α is maximum angle for which cardboard strip does not topple then find the value of $9 \tan^2 \alpha$? (Assume that it does not slide).



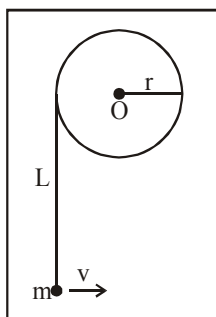
27. A pencil having a cross section of a regular n -side polygon of side a is kept on an incline plane of inclination 22.5° with horizontal. Figure represents the situation for $n = 6$. For what maximum value of n pencil will not topple if friction is sufficient to prevent slipping ?



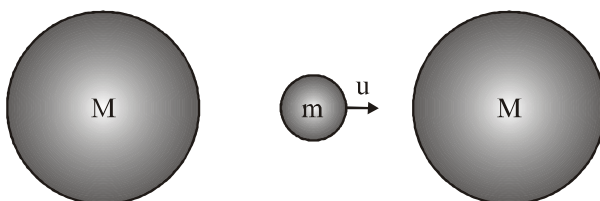
28. A coordinate axis system taking x-axis as horizontal smooth floor is shown in figure. Two small balls of masses 'm' and '3 m' attached with a string are released from some heights on y-axis as shown in figure. The balls may collide head on or obliquely. After a certain time, mass 'm' is at (9 cm, 20 cm) while mass '3 m' is 25 cm above the x axis and the string is taut. The balls always remain in x-y plane. Then the length (in cm) of string is.



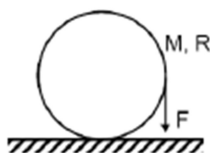
29. A horizontal disk of radius $r = 0.2$ m is fixed onto a horizontal frictionless table. One end of a massless string of length $L = 0.8$ m is fixed to the perimeter of the disk, while the other end is attached to an object of mass $m = 0.6$ kg, which stands on the table as shown. The object is then given a velocity of magnitude $v = 0.4$ m/s in a direction perpendicular to the string. Then the time (in s) after which object will hit disc, is:



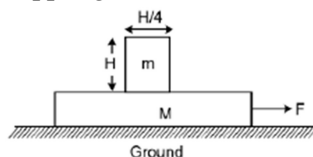
30. Two men A and B, A standing at periphery of circle of radius R and B standing at centre of the circle. A starts moving on the periphery with constant tangential acceleration and B starts moving with constant velocity v at an angle θ , from the line joining initial position of A and centre, such that they meet during subsequent motion, for this purpose maximum possible value of v is $\sqrt{\frac{aR}{\alpha\theta}}$. here α is an integer. Find α .
31. Two identical uniform spheres of mass M are at rest and another uniform sphere of mass m is moving along their line of centers between them with speed u . After second collision middle sphere move with velocity $0.36 u$. If spheres are perfectly elastic, then calculate $\frac{M}{m}$.



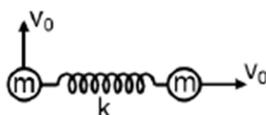
32. Moment of inertia of a disc of mass $M = 2 \pm 0.02$ kg, inner radius $a = 5 \pm 0.05$ cm, outer radius $b = 7 \pm 0.07$ cm, with respect to an axis passing through geometrical centre of the disc and perpendicular to the plane of disc is given by $(A_0 \pm B_0 \times 10^{-2}) \text{ kg} - \text{cm}^2$. Find the value of $\frac{B_0}{A_0}$.
33. A circular disc of mass M is placed over a rough horizontal surface. Coefficient of friction is μ . A force F is applied on the periphery of the disc, tangentially in vertically downward direction. The disc will undergo pure rolling only if $F \leq F_0$. When $F = F_0$, the value of $\frac{(2-3\mu)F_0}{\mu Mg}$ is :



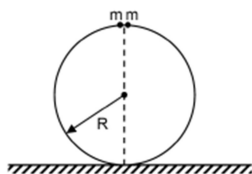
34. A block of mass $m = 2$ kg of shown dimensions is placed on a plank of mass $M = 6$ Kg which is placed on smooth horizontal plane. The coefficient of friction between the block and the plank is $\mu = \frac{1}{3}$. If a horizontal force F is applied on the plank, then find the maximum value of F (in N) for which the block and the plank move together without toppling the block.



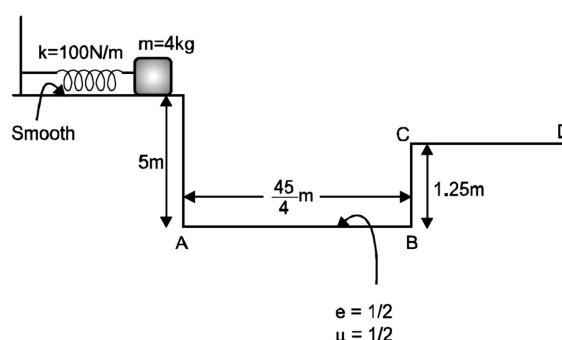
35. Two small spheres of mass 1 kg each are connected to each other by means of an undeformed spring of force constant $k = 100$ N/m and natural length 1 m. The system is placed on a smooth horizontal surface and the two spheres are given horizontal velocities v_0 along and perpendicular to the spring, as shown in the figure. The maximum elongation in the spring is found to be 0.1 m during the subsequent motion. The value of v_0 is α m/s. Find the integer next to α .



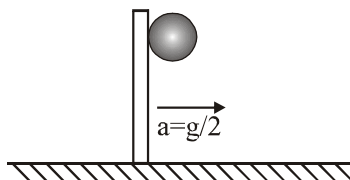
36. Two beads each of mass ' m ' are positioned at the top of a frictionless hoop of mass M and radius R , which stands vertically on ground. The beads are pushed gently so that they slide down the hoop, one to the right and one to the left. If during subsequent motion hoop is to lose contact from ground, then minimum value of $\frac{m}{M}$ is α . Find 4α .



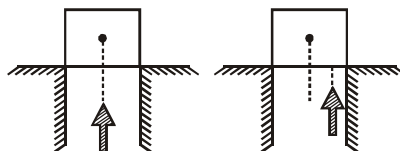
37. A man crosses a river of width d . Current flow speed is v . Speed of swimmer relative to water is v . Man always heads towards the point exactly opposite to the starting point at the another bank (relative to water). If radius of curvature of the path followed by the swimmer just after he start swimming is $n\sqrt{2}d$, then the value of n is.
38. A small block is released from rest when compression in spring is x . At natural length, spring detaches from block. Collision of block with rough surface AB ($\mu = \frac{1}{2}$) is inelastic ($e = \frac{1}{2}$). Find minimum value of x for which block can land on surface CD (neglect rotation of block during entire motion and dimensions of blocks are very small). [$g = 10 \text{ m/s}^2$] :-



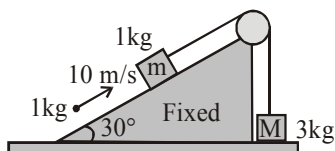
39. A thin uniform rod of mass 0.25 kg and length 1.20 m is smoothly hinged about a point passing through its centre of mass. Its angular momentum as a function of time t is given by $L = \frac{2}{3}t^3 + 2t^2$ (where t is in sec. and L is in $\text{kg m}^2/\text{sec.}$). Angular acceleration of the rod at $t = \frac{3}{2} \text{ sec.}$ is $50 \text{ N (in rad/sec}^2\text{)}$, then find 'N'.
40. A uniform solid sphere rolls down on a vertical rigid surface without slipping. The vertical surface moves with a constant acceleration $a = g/2$ in horizontal direction as shown. The minimum coefficient of friction between the sphere and vertical surface so as to prevent slipping is μ . Write 14μ as answer.



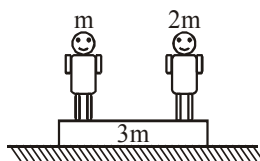
41. Two identical blocks are resting on supports as shown. Two identical bullets travelling with the same speed hit the blocks & get embedded in them in very short time. The blocks leave the contact immediately after bullet strikes them. In case (i) bullet's velocity is toward centre of mass and in case (ii) bullet's velocity is off centre. The kinetic energies after collision are K_1 & K_2 and maximum height reached by the centre of mass are h_1 & h_2 respectively. then :



- (A) $K_1 > K_2$ (B) $K_1 = K_2$ (C) $h_1 < h_2$ (D) $h_1 = h_2$
42. A light inextensible thread passes over a small frictionless pulley. Two blocks of masses $m = 1$ kg and $M = 3$ kg respectively are attached with the thread as shown in the figure. The heavier block rests on a horizontal surface. A shell of mass 1 kg moving upwards (along the smooth inclined plane) with a velocity 10 m/s collides perfectly in-elastically with the block of mass 'm' at $t = 0$. Then

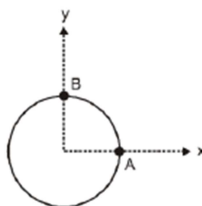


- (A) Maximum height ascended by M is $\frac{1}{4}$ m.
 (B) String remains slacked for time 1 s.
 (C) Maximum height reached by M is at $t = 2.5$ s.
 (D) There is no turning in the motion of M.
43. In the given figure, surface between plank and ground is smooth. Initially both men are at rest at the ends of the plank. Now man of mass m starts to move with constant relative acceleration 'a' with respect to plank towards man of mass $2m$. $2m$ doesn't move with respect to plank. Then choose the correct option (length of plank is ℓ) -



- (A) Friction force between $2m$ and plank is $\frac{ma}{3}$.
 (B) Men will meet after time $\sqrt{\frac{12\ell}{7a}}$.
 (C) Net force on plank is $\frac{ma}{4}$.
 (D) Friction force between man of mass m and plank is ma .

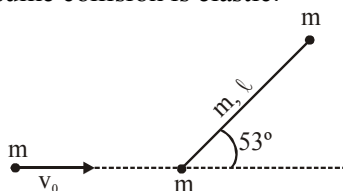
44. A uniform disc of mass m and radius R , hinged at some point is performing pure rotation with respect to hinge, in horizontal plane with a constant angular velocity ω . At certain instant, center of the disc is at origin in the mentioned co-ordinate system and velocity of particle A is $\vec{V}_A = -\frac{\omega R}{4}(3\hat{i} - 4\hat{j})$ m/s. Kinetic energy of the disc is :



- (A) $\frac{mR^2\omega^2}{2}$ (B) $\frac{17mR^2\omega^2}{4}$ (C) $\frac{17mR^2\omega^2}{32}$ (D) $\frac{7mR^2\omega^2}{32}$

Paragraph for questions 45 and 46

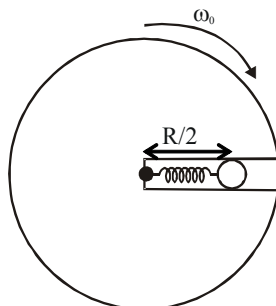
A rigid rod has length ℓ and mass m . Two point object each of same mass m is attached to the ends of rod. The rod is kept on a smooth horizontal surface. The rod is struck by a particle of mass m moving with velocity v_0 as shown in figure. Assume collision is elastic.



45. Velocity of the particle just after the collision is :
 (A) $\frac{113}{422}v_0$ (B) $\frac{103}{422}v_0$ (C) $\frac{103}{437}v_0$ (D) $\frac{113}{437}v_0$
46. Angular velocity of rod just after the collision is :
 (A) $\frac{1184v_0}{1477\ell}$ (B) $\frac{1284v_0}{1477\ell}$ (C) $\frac{180v_0}{211\ell}$ (D) $\frac{160v_0}{211\ell}$

Paragraph for Question Nos. 47 to 48

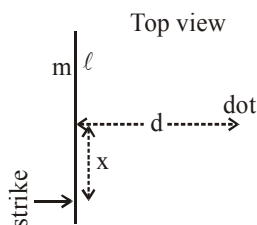
A uniform disc of mass ' m ' and radius ' R ' is free to rotate in horizontal plane about a vertical smooth fixed axis passing through its centre. There is a smooth groove of small width along the radius of the disc and one small ball of same mass is attached to a spring as shown in figure. Initially spring is in its natural length of $\frac{R}{2}$. The disc is given initial angular velocity ω_0 and released (spring constant $k = \frac{m\omega_0^2}{2}$)



47. Angular velocity of disc when elongation in spring becomes $\frac{R}{2}$, will be :
- (A) $\frac{\omega_0}{2}$ (B) ω_0 (C) $\frac{\omega_0}{4}$ (D) $\frac{3\omega_0}{4}$
48. Radial velocity of ball when elongation in spring becomes $\frac{R}{2}$, will be :
- (A) $\omega_0 R$ (B) $\frac{\omega_0 R}{2}$ (C) $\frac{\omega_0 R}{4}$ (D) $\frac{3\omega_0 R}{4}$

Paragraph for Question Nos. 49 to 50

A thin uniform rod of length ℓ and mass m lies on a frictionless fixed horizontal surface. It is struck with a quick horizontal blow (directed perpendicular to the rod) at a distance 'x' from the center of rod. A dot is painted on the horizontal surface at a distance d from the initial position of the center of the rod.



49. Let v be the speed of centre of mass and ω be the angular speed of the rod just after the blow. Then the value of $\frac{\omega}{v}$ is :
- (A) $\frac{x}{\ell^2}$ (B) $\frac{3x}{\ell^2}$ (C) $\frac{4x}{\ell^2}$ (D) $\frac{12x}{\ell^2}$
50. If the rod makes one revolution by the time centre reaches the dot, then by adjusting 'x', minimum possible value of 'd' (in terms of ℓ) should be :
- (A) $\frac{\pi\ell}{2}$ (B) $\frac{\pi\ell}{3}$ (C) $\frac{\pi\ell}{4}$ (D) $\frac{\pi\ell}{6}$

ANSWER KEY TO DPP # 04 (JEE-ADVANCED)

1. (A,C)	2. (A,B,C,D)	3. (B,C)	4. (A,B,C)	5. (B,C,D)
6. (A,B,C)	7. (B,C,D)	8. (A,C)	9. (A,B)	10. (A,D)
11. (A,B,C,D)	12. (A, B,C,D)	13. (B,C)	14. (A,B,C,D)	15. (A,B,C)
16. (A,C)	17. (A,C)	18. (A,C,D)	19. (A,B,C,D)	
20. (A,B,C,D)	21. 30	22. 3	23. 4	24. 4
25. 5	26. 4	27. 8	28. 13	29. 4
30. 2	31. 4	32. 3	33. 3	34. 20
35. 2	36. 6	37. 2	38. 2	39. 7
40. 8	41. (D)	42. (C)	43. (A)	44. (C)
45. (B)	46. (C)	47. (A)	48. (B)	49. (D)
50. (B)				